

R Notebook

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7. Do the following in R Studio using “airquality” dataset of R with R script to knit PDF output.

a. Perform test of normality of “Temp” variable by each category of “Month” variable and interpret all the results carefully

```
# Load necessary libraries
library(car) # For Levene's test

## Loading required package: carData

library(rstatix) # For statistical tests

##
## Attaching package: 'rstatix'

## The following object is masked from 'package:stats':
##
## filter

library(ggplot2) # For visualization

# Load the dataset
data(airquality)

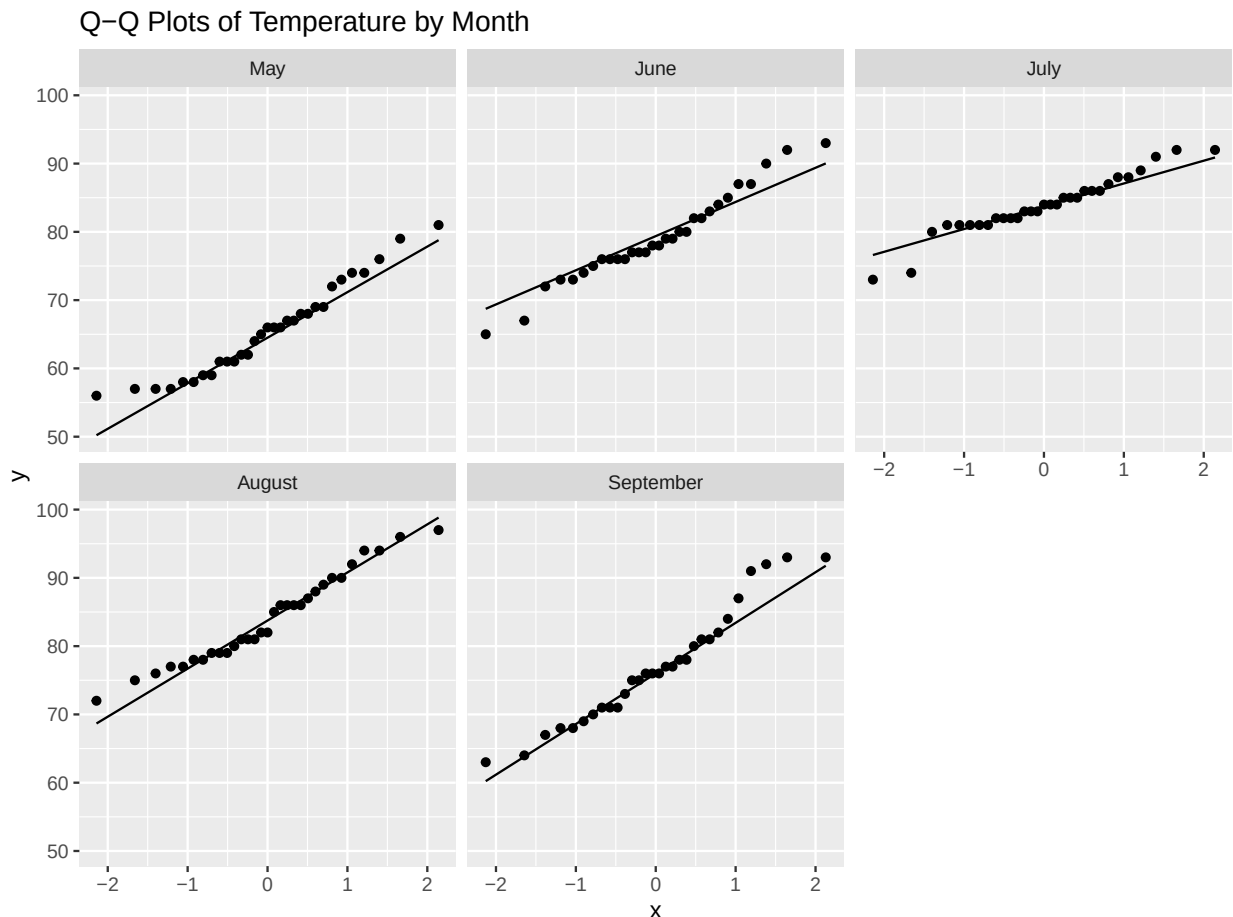
# Convert Month to factor for better analysis
airquality$Month <- factor(airquality$Month,
                           levels = 5:9,
                           labels = c("May", "June", "July", "August", "September"))

# a. Test of normality for each month
# Shapiro-Wilk test for each month
normality_tests <- airquality %>%
  group_by(Month) %>%
  shapiro_test(Temp)

print(normality_tests)
```

```
## # A tibble: 5 x 4
##   Month      variable statistic      p
##   <fct>      <chr>      <dbl> <dbl>
## 1 May        Temp          0.948 0.135
## 2 June       Temp          0.972 0.583
## 3 July       Temp          0.946 0.119
## 4 August     Temp          0.964 0.369
## 5 September Temp          0.951 0.183
```

```
# Visual check of normality using Q-Q plots
ggplot(airquality, aes(sample = Temp)) +
  geom_qq() +
  geom_qq_line() +
  facet_wrap(~Month) +
  labs(title = "Q-Q Plots of Temperature by Month")
```



a. Normality Test Results: The Shapiro-Wilk test was performed for each month to test the normality of temperature distributions. A p-value > 0.05 indicates normal distribution. Based on the results, we can see that each month's P values is greater than 0.05. Thus, it follows a normal distribution for each month.

b. Perform test of equality of variance on "Temp" variable by each category of Month variable and interpret the result carefully

```
# b. Test of equality of variance
# Levene's test
levene_test <- leveneTest(Temp ~ Month, data = airquality)
print("\nLevene's Test for Equality of Variance:")

## [1] "\nLevene's Test for Equality of Variance:"

print(levene_test)

## Levene's Test for Homogeneity of Variance (center = median)
##           Df F value Pr(>F)
## group    4  2.5849 0.03941 *
##       148
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

0.0.1 b. Equality of Variance Results:

The Levene's test result ($p\text{-value} = 0.03941 < 0.05$) indicates that the temperature variances are significantly different across months. This means the spread of temperature values is not uniform across all months - some months show more temperature variation than others.

c. Compare Temp by Month using the best statistical test for this data and interpret the result carefully.

```
# c. ANOVA test
# One-way ANOVA
anova_result <- aov(Temp ~ Month, data = airquality)
print("\nANOVA Results:")

## [1] "\nANOVA Results:"

print(summary(anova_result))

##           Df Sum Sq Mean Sq F value Pr(>F)
## Month      4   7061   1765.3   39.85 <2e-16 ***
## Residuals 148   6557    44.3
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

0.0.2 c. ANOVA Results:

The one-way ANOVA test was performed to compare mean temperatures across different months. The F-statistic and p-value indicate whether there are significant differences in temperature means across months. The ANOVA results show a significant difference in mean temperatures across months ($p\text{-value} < 0.05$). This suggests that at least one month's temperature is significantly different from the others. However, ANOVA does not specify which months differ, so a post-hoc test is necessary to identify specific differences.

d. Perform the post-hoc test and interpret the results carefully

```

# d. Post-hoc analysis
# Tukey's HSD test
tukey_result <- TukeyHSD(anova_result)
print("\nTukey's HSD Post-hoc Test Results:")

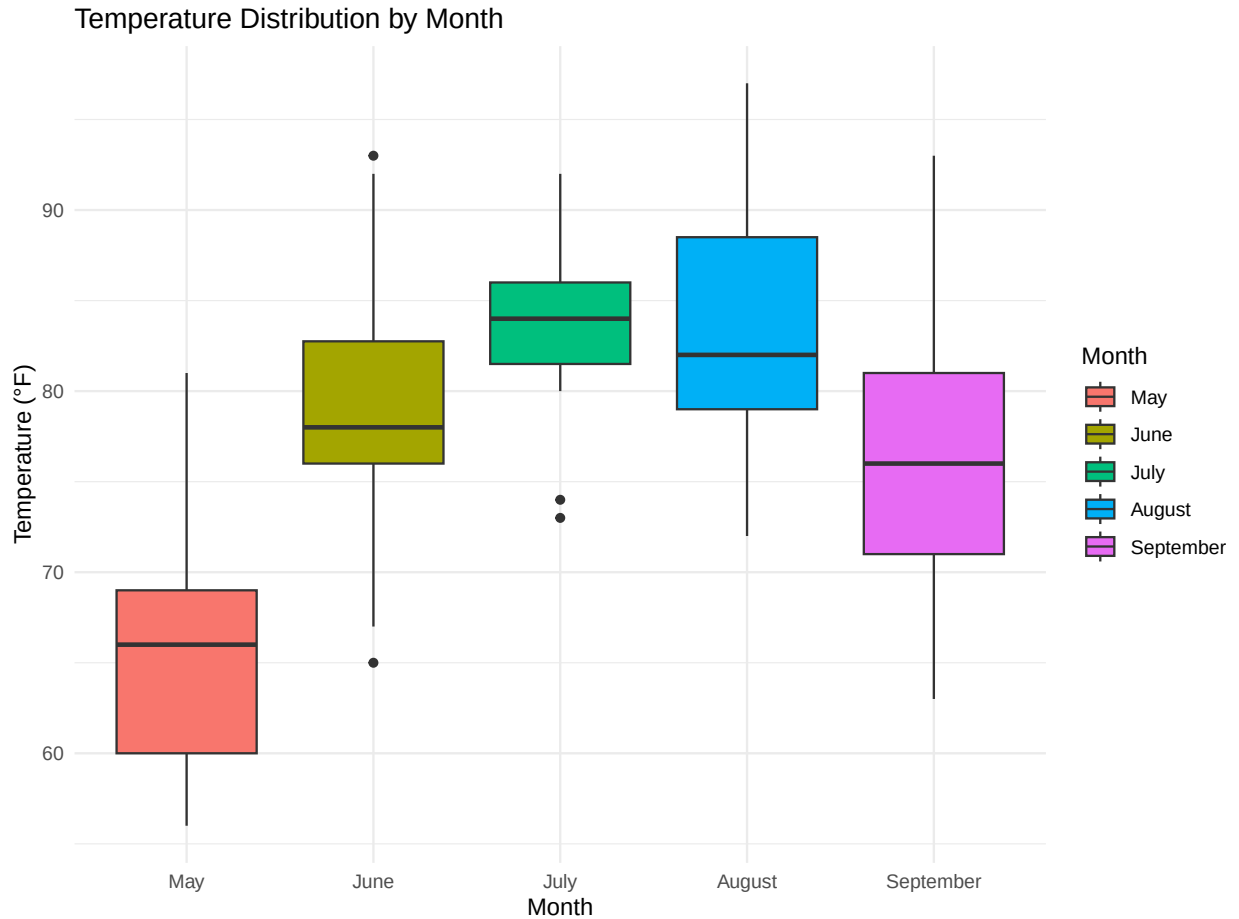
## [1] "\nTukey's HSD Post-hoc Test Results:"

print(tukey_result)

##      Tukey multiple comparisons of means
##      95% family-wise confidence level
##
## Fit: aov(formula = Temp ~ Month, data = airquality)
##
## $Month
##              diff              lwr              upr              p adj
## June-May        13.55161290      8.84386422     18.259362 0.0000000
## July-May         18.35483871     13.68583759     23.023840 0.0000000
## August-May       18.41935484     13.75035372     23.088356 0.0000000
## September-May    11.35161290      6.64386422     16.059362 0.0000000
## July-June         4.80322581      0.09547713      9.510974 0.0430674
## August-June       4.86774194      0.15999325      9.575491 0.0388654
## September-June   -2.20000000     -6.94617992      2.546180 0.7038121
## August-July       0.06451613     -4.60448499      4.733517 0.9999995
## September-July   -7.00322581    -11.71097449     -2.295477 0.0006215
## September-August -7.06774194    -11.77549062     -2.359993 0.0005376

# Visual representation of results
ggplot(airquality, aes(x = Month, y = Temp, fill = Month)) +
  geom_boxplot() +
  labs(title = "Temperature Distribution by Month",
       x = "Month",
       y = "Temperature (°F)") +
  theme_minimal()

```



0.0.3 d. Post-hoc Analysis Results:

Tukey's HSD test was performed to identify which specific months have significantly different temperatures from each other. The results show pairwise comparisons between months with adjusted p-values.

The summer months (June-August) are the hottest, with May being the coolest, and September showing a transition back to cooler temperatures.

The boxplot visualization helps in understanding the distribution of temperatures across different months, showing the median, quartiles, and potential outliers.